

06_Class_Activity

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In class activity 6:

What did we do last time in activity 5?

- Understanding standard normal distributions and z-scores
- Calculating and interpreting standard error
- Creating confidence intervals
- Working with the Student's t-distribution

Today's focus:

- Review more r code
- understand α alpha and β beta errors
- do more
 - 1 sample t tests
 - 2 sample t tests

```
# Install packages if needed (uncomment if necessary)
# install.packages("readr")
# install.packages("tidyverse")
# install.packages("car")
# install.packages("here")

# Load libraries
library(patchwork)
library(car)           # For diagnostic tests
library(tidyverse)     # For data manipulation and visualization
library(readxl)

# Load the pine needle data
# Use here() function to specify the path
pine_switch_df <- read_excel("data/class_pine needle length switched.xlsx")

# Examine the first few rows
head(pine_switch_df)
```

```
# A tibble: 6 × 5
  group tree_no tree_char side length_mm
<chr>   <dbl> <chr>    <chr>    <dbl>
1 five     1 tree_1 sunny     22.7
2 five     1 tree_1 sunny     21.1
3 five     1 tree_1 sunny     18.6
4 five     1 tree_1 sunny     18.6
5 five     1 tree_1 sunny     21.0
6 five     1 tree_1 sunny     18.9
```

```
ps_df <- pine_switch_df %>%
  group_by(group, tree_no, tree_char, side) %>%
  summarise(length_mm = mean(length_mm, na.rm=TRUE))
```

`summarise()` has grouped output by 'group', 'tree_no', 'tree_char'. You can override using the `.groups` argument.

```
ps_shady_df <- ps_df %>%
  filter(side == "shady")

ps_sunny_df <- ps_df %>%
  filter(side == "sunny")

head(ps_df)
```

```
# A tibble: 6 × 5
# Groups:   group, tree_no, tree_char [3]
  group          tree_no tree_char side length_mm
<chr>          <dbl> <chr>   <chr>   <dbl>
1 big_fat_fecund_female_fish      2 tree_2  shady    15.4
2 big_fat_fecund_female_fish      2 tree_2  sunny    13.2
3 bill                          3 tree_3  shady    16.7
4 bill                          3 tree_3  sunny    16.0
5 ciabatta                      5 tree_5  shady    19.1
6 ciabatta                      5 tree_5  sunny    17.7
```

Part 1: Exploratory Data Analysis

Before conducting hypothesis tests, we should always explore our data to understand its characteristics.

Let's calculate summary statistics and create visualizations.

Activity: Calculate basic summary statistics for pine needle length

```
# YOUR TASK: Calculate summary statistics for pine needle length
# Hint: Use summarize() function to calculate mean, sd, n, etc.

# Create a summary table for all pine needles
pine_summary <- ps_df %>%
  group_by(side) %>%
  summarize(
    mean_length = mean(length_mm, na.rm=TRUE),
    sd_length = sd(length_mm, na.rm=TRUE),
    n = sum(!is.na(length_mm)),
    se_length = sd_length / (n^0.5),
    t_critical = qt(0.975, df = n - 1), # 95% CI uses 0.975 (two-tailed)
    ci_lower = mean_length - t_critical * se_length,
    ci_upper = mean_length + t_critical * se_length
  )

print(pine_summary)
```

```
# A tibble: 2 × 8
  side mean_length sd_length n se_length t_critical ci_lower ci_upper
<chr>    <dbl>    <dbl> <int>    <dbl>    <dbl>    <dbl>    <dbl>
1 shady      17.6      2.51     8     0.886      2.36     15.5     19.7
2 sunny      16.2      2.64     8     0.934      2.36     13.9     18.4
```

```
# Now calculate summary statistics by wind exposure
# YOUR CODE HERE
```

Part 1: Visualizing the Data

Activity: Create visualizations of pine needle length

Create a histogram and a boxplot to visualize the distribution of pine needle length values.

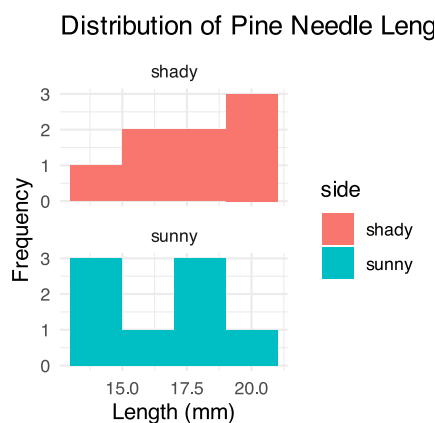
Effective data visualization helps us understand:

- The central tendency
- The spread of the data
- Potential outliers
- Shape of distribution

Your Task

```
# YOUR TASK: Create a histogram of pine needle length
# Hint: Use ggplot() and geom_histogram()

# Histogram of all pine needle lengths
ggplot(ps_df, aes(x = length_mm, fill = side)) +
  geom_histogram(binwidth = 2) +
  labs(title = "Distribution of Pine Needle Lengths",
       x = "Length (mm)",
       y = "Frequency") +
  theme_minimal() + facet_wrap("side", ncol = 1)
```



```
# how can you do this by side to see both plots
```

What is the Effect size or difference in means?

💡 Practice Exercise: Calculate Effect size

We could also look at the difference in means... some cool code here

```
# Assuming your dataframe is called df
pine_summary %>%
  summarize(difference = mean_length[side == "sunny"] - mean_length[side == "shady"])
```

```
# A tibble: 1 × 1
  difference
    <dbl>
1      -1.45
```

Part 2: Conducting the Two-Sample T-Test

Activity: Conduct a two-sample t-test

Now we can compare the mean pine needle lengths between sunny and shady sides.

$H_0: \mu_1 = \mu_2$ (The mean needle lengths are equal)

$H_1: \mu_1 \neq \mu_2$ (The mean needle lengths are different)

Deciding between:

- Standard t-test (equal variances)
- Welch's t-test (unequal variances)

Based on our Levene's test result.

```
# YOUR TASK: Conduct a two-sample t-test
# Use var.equal=TRUE for standard t-test or var.equal=FALSE for Welch's t-test

# Standard t-test (if variances are equal)
t_test_result <- t.test(length_mm ~ side, data = ps_df, var.equal = TRUE)
print("Standard two-sample t-test:")
```

```
[1] "Standard two-sample t-test:"
```

```
print(t_test_result)
```

Two Sample t-test

```
data: length_mm by side
t = 1.1279, df = 14, p-value = 0.2783
alternative hypothesis: true difference in means between group shady and group sunny is not
equal to 0
95 percent confidence interval:
 -1.309330  4.214005
sample estimates:
```

mean in group shady	mean in group sunny
17.60561	16.15328

What is Power

Statistical power represents the probability of detecting a true effect (rejecting the null hypothesis when it is false). In this case, with a power of 97%, there's a 97% chance of detecting a true difference of 30 units between the means of the two groups if such a difference actually exists.

A power analysis like this is typically done for one of these purposes:

1. Before data collection to determine required sample size
2. After a study to evaluate if the sample size was adequate
3. To determine the minimum detectable effect size with the given sample

With 97% power, this test has excellent ability to detect the specified effect size. Generally, **80% power is considered acceptable**, so 97% indicates a very well-powered study for detecting a difference of 30mm between the groups.

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

```
# Calculate power for detecting a 1 mm difference
side_diff <- 1

# Get sample sizes # note assumes no missing values
sunny_n <- nrow(ps_sunny_df)
shady_n <- nrow(ps_shady_df)

# Calculate pooled standard deviation
sun_sd_pooled <- sqrt((var(ps_sunny_df$length_mm) * (sunny_n - 1) +
                      var(ps_shady_df$length_mm) * (shady_n - 1)) /
                      (sunny_n + shady_n - 2))

# Calculate Cohen's d effect size
sun_effect_size <- side_diff / sun_sd_pooled

# Calculate degrees of freedom
sun_df <- sunny_n + shady_n - 2

# Set significance level
sun_alpha <- 0.05

# Calculate power (fixed parameters)
sun_power <- power.t.test(n = min(sunny_n, shady_n),
                          delta = side_diff, # Raw difference, not effect size
                          sd = sun_sd_pooled, # Use pooled SD, not side_diff
                          sig.level = sun_alpha, # Use 0.05, not 0.5
                          type = "two.sample",
                          alternative = "two.sided")

# Display results
print("Sample sizes:")
```

```
[1] "Sample sizes:"
```

```
print(paste("Sunny:", sunny_n, "Shady:", shady_n))
```

```
[1] "Sunny: 8 Shady: 8"
```

```
print(paste("Pooled SD:", round(sun_sd_pooled, 3)))
```

```
[1] "Pooled SD: 2.575"
```

```
print(paste("Effect size (Cohen's d):", round(sun_effect_size, 3)))
```

```
[1] "Effect size (Cohen's d): 0.388"
```

```
print("")
```

```
[1] ""
```

```
print("Power analysis results:")
```

```
[1] "Power analysis results:"
```

```
print(sun_power)
```

Two-sample t test power calculation

```
      n = 8
  delta = 1
     sd = 2.575237
sig.level = 0.05
  power = 0.1083557
alternative = two.sided
```

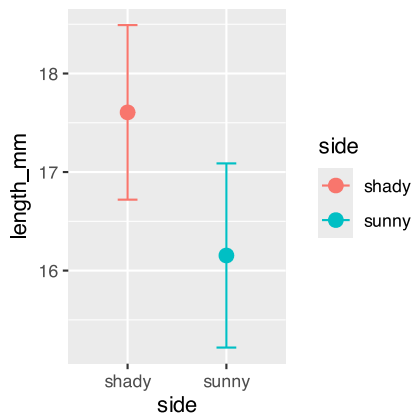
NOTE: n is number in *each* group

Now to make a final plot

Typically we will make a plot that has the mean and standard error on it to represent the data

Your task is to make a more publication quality final plot

```
pine_mean_se <- ps_df %>%
  ggplot(aes(side, length_mm, color = side))+
  stat_summary(fun = "mean", na.rm=TRUE, geom="point", size = 3)+
  stat_summary(fun.data = "mean_se", width = 0.2, geom = "errorbar")
pine_mean_se
```



Now make a new theme

This is how you can customize a plot with your own theme

Run this block or ctrl/command + enter on the very start of the code as running inside the function fails.

```
# Load your custom theme -----
theme_class <- function(base_size = 14, base_family = "Sans")
{theme(
  # REMOVE PLOT FILL AND GRIDS
  panel.background=element_rect(fill = "transparent", colour = "transparent"),
  plot.background=element_rect(fill="transparent", colour=NA),
  # removes the grid lines
  panel.grid.major = element_line(linetype = "blank"),
  panel.grid.minor = element_line(linetype = "blank"),
  # X and Y LABELS APPEARANCE
  axis.text = element_text(colour = "black"), # this is for the text on ticks and
  overrides all below if not set
  axis.title.x=element_text(size=18, face="bold"), # this is for the x title
  axis.title.y=element_text(size=18, face="bold"), # this is for the y title
  axis.text.x = element_text(size=16, face="bold", angle=0, vjust = .5, hjust=.5), # for X
  tick labels
  axis.text.y = element_text(size=16, face="bold"),# for Y tick labels
  # plot.title = element_text(hjust = 0.5, colour="black", size=22, face="bold"),
  # ADD AXES LINES AND SIZE
  axis.ticks = element_line(colour = "black"),
  axis.line.x = element_line(color="black", linewidth = 0.5, linetype = "solid" ),
  axis.line.y = element_line(color="black", linewidth = 0.5, linetype = "solid"),
  # LEGEND
  # LEGEND TEXT
  legend.text = element_text(colour="black", size = 16, face = "bold"),
  # LEGEND TITLE
  legend.title = element_text(colour="black", size=18, face="bold"),
  # LEGEND POSITION AND JUSTIFICATION
  # legend.justification=c(0.1,1),
  legend.position="right",
  # REMOVE BOX BEHIND LEGEND SYMBOLS
  legend.key = element_rect(fill = "transparent", colour = "transparent"),
  # REMOVE LEGEND BOX
  legend.background = element_rect(fill = "transparent", colour = "transparent"),
  # ADD PLOT BOX
  # panel.border = element_rect(color = "black", fill = NA, size=2),
  # turn off an element element_blank()
```

```

panel.border = element_blank(),
plot.title = element_text(hjust = 0, vjust=2.12),
plot.caption = element_text(hjust = 0, vjust=1.12))
}

```

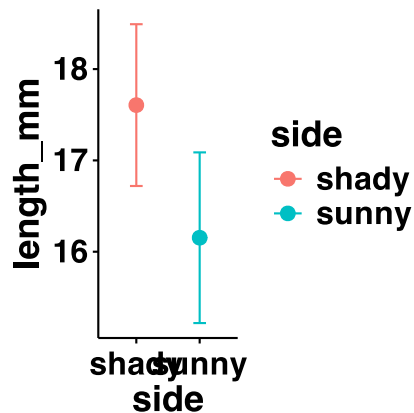
Now apply your theme to the plot above

We can add things to an existing plot as we go.. and it is not over writing the plot as we are not using the <-

```

pine_mean_se +
  theme_class()

```

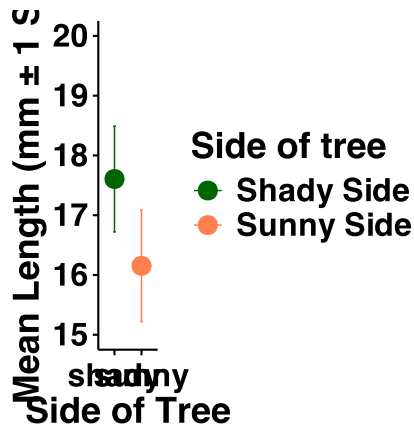


Further modifications

```

pine_mean_se <- ps_df %>%
  ggplot(aes(side, length_mm, color = side))+
  stat_summary(fun = "mean", na.rm=TRUE, geom="point",
              size = 4)+
  stat_summary(fun.data = "mean_se", geom = "errorbar",
              width = 0.1, linewidth = 0.35) +
  labs(x = "Side of Tree",
       y = "Mean Length (mm \u00B1 1 SE))+
  coord_cartesian(ylim = c(15, 20))+
  theme_class()+
  scale_color_manual(
    name = "Side of tree",
    labels = c(
      "shady" = "Shady Side",
      "sunny" = "Sunny Side"),
    values = c(
      "shady" = "darkgreen",
      "sunny" = "coral")
  )
pine_mean_se

```

Summary and Conclusions

In this activity, we've:

1. Formulated hypotheses about pine needle length
2. Tested assumptions for parametric tests
3. Conducted a two-sample t-tests
4. Visualized data using appropriate methods
5. Learned how to interpret and report t-test results

Key takeaways:

- Always check assumptions before conducting tests
- Visualize your data to understand patterns
- Report results comprehensively
- Consider alternatives when assumptions are violated

Reflection Questions

After completing the activities, discuss these questions with your group:

1. How does sample size affect our confidence in estimating the population mean?
2. Why is the t-distribution more appropriate than the normal distribution when working with small samples?
3. When comparing two populations, what can we learn from looking at confidence intervals versus performing a t-test?
4. How would you explain the concept of statistical significance to someone who has never taken a statistics course?
5. What do we do if assumptions FAIL!!!